



COMP9444: Neural Networks and Deep Learning

Week 1a. Overview

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Course Web Pages

- ➔ <https://www.cse.unsw.edu.au/~cs9444/25T2/>
- ➔ <https://webcms3.cse.unsw.edu.au/COMP9444/25T2/>

Lecturers, Course Admins

- ➔ Alan Blair (Lecturer-in-Charge)
- ➔ Hao Xue (Lecturer)
- ➔ Zahra Donyavi (Course Admin)
- ➔ Fatemeh Morgani (Course Admin)
- ➔ cs9444@cse.unsw.edu.au

AlphaGo



A woman plays Go with an AI-powered robot developed by the firm SenseTime, based in Hong Kong. Credit: Joan Cros/NurPhoto via Getty



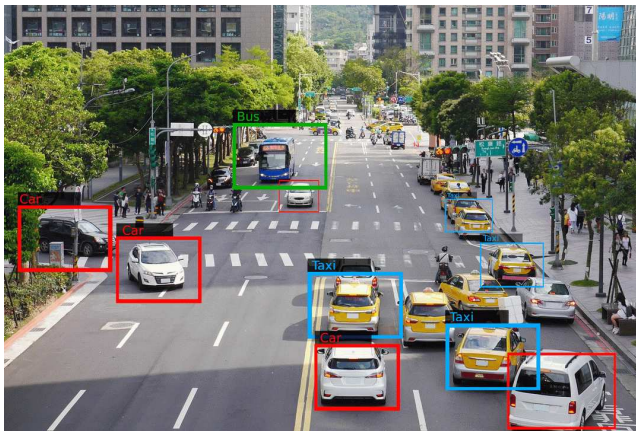
AlphaFold – Nobel Prize in Chemistry 2024



AlphaFold, a deep learning algorithm that predicts a protein's 3D structure.

Source: <https://deepmind.google/technologies/alphafold/>

Object Detection



Source: https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/Redmon_You_Only_Look_CVPR_2016_paper.pdf

MultiModal AI – Image Captioning



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



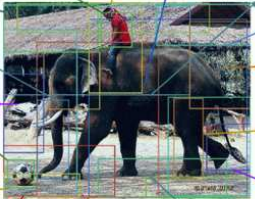
"two young girls are playing with lego toy."

Source: <https://cs.stanford.edu/people/karpathy/deepimagesent/>

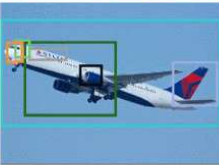
Dense Image Captioning



people are in the background
light on the wall
sign on the wall
man wearing a white shirt
man sitting on a table
man wearing blue jeans
blue jeans on the ground
man sitting on a bench
man sitting on a table
man wearing black shirt
white laptop on a table
man sitting on a table
woman wearing a black shirt
chair is brown
floor is brown



man wearing a black shirt
red shirt on a man
elephant is standing
elephant is brown
roof of a building
green trees in the background
leg of an elephant
leg of an elephant
shadow on the ground
elephant is standing
ground is brown
ground is visible
ball is white
rocks on the ground
trunk of an elephant
large green trees



Our Model: plane is flying, tail of the plane, red and white plane, plane is white, engine on the plane, windows on the plane, nose of the plane.

Full Image RNN: A large jetliner flying through a blue sky.



woman wearing a black shirt, teddy bear is brown, chair is black, glass of wine, table is brown, woman with brown hair, paper on the table.

Full Image RNN: A man and a woman sitting at a table with a cake.



teddy bear is wearing a red shirt, red and white teddy bear, bear is wearing a red hat, red and white shirt, table is brown, black nose of a bear.

Full Image RNN: A teddy bear with a red bow on it.



train on the tracks, trees are green, front of the train is yellow, grass is green, green trees in the background, photo taken during the day, red train car.

Full Image RNN: A train is traveling down the tracks near a forest.

Source: <https://cs.stanford.edu/people/karpathy/densecap/>

Visual Question Answering

Who is wearing glasses?

man



woman

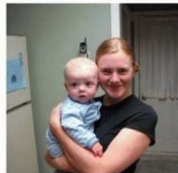


Where is the child sitting?

fridge



arms



Is the umbrella upside down?

yes



no



How many children are in the bed?

2



1



Source: <https://visualqa.org>

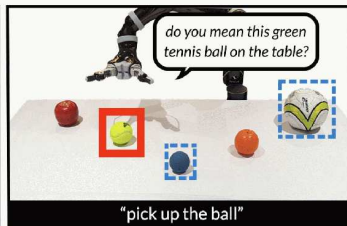
Visual Grounding



(a)



(b)



(c)

Source: <https://drive.google.com/file/d/15AttCp-KCDEt8Ys5TfqXowsElm9GqAkH/view?pli=1>

Recent Advances in AI and Deep Learning

- Game Playing
- Medical Research and Development
- Text Generation, ChatBots, Code Generation
- Image, Voice, Song, Video Generation
- Object Detection, Image Segmentation and Captioning, Multimodal AI
- Agents, Decision-Making AI

Artificial Intelligence, Machine Learning, Deep Learning

- **Artificial Intelligence:** Development of smart systems and machines to carry out tasks which traditionally required human intelligence.
- **Machine Learning:** Create algorithms that can learn from data and make decisions based on patterns observed. Requires human intervention when decision is incorrect.
- **Deep Learning:** Uses complex and deep artificial neural networks to reach accurate conclusions without human intervention. Typically requires large-scale annotated data for training.

Lecture / Tutorial / Lab Schedule

- Lectures (Weeks 1-5, 7-10, Science Theatre K-F13-G09)
 - Tuesday 2-4
 - Wednesday 11-1
- Tutorials (Weeks 1-5)
- Group Mentoring Sessions (Weeks 7-10)

Teaching Strategies

- ➔ Lecture Slides and Tutorial Questions will be delivered through the course WebCMS site.
- ➔ Lecture Recordings, Forums, Group Formation and MyExperience will be delivered through the course Moodle site.
- ➔ Lecture time will be used to summarize the material, discuss recent developments, and answer questions.
- ➔ Tutorials in Weeks 1 to 5, to discuss worked examples and develop a deeper understanding of fundamental topics.
- ➔ Mentoring Sessions (Weeks 7 to 10) to assist with Group Project.

Teaching Strategies

You must keep up with lectures, either by attending in person or watching the recordings. Students enrolled in the Web stream are welcome to attend in person if space is available.

You are expected to:

- ➔ review course materials before and after each lecture
- ➔ attempt tutorial questions beforehand and be ready to ask questions
- ➔ complete tutorial exercises, assignment and group project
- ➔ discuss the material with your fellow students if possible
- ➔ consider further exploring topics of particular interest
- ➔ ask questions and contribute to discussion in online Moodle forums

Textbook(s)

Understanding Deep Learning

by Simon J.D. Prince

MIT Press, 2023

<https://udlbook.github.io/udlbook/>

Deep Learning

by Ian Goodfellow, Yoshua Bengio and Aaron Courville

MIT Press, 2014

<http://www.deeplearningbook.org>

<https://mitpress.mit.edu/books/deep-learning>

Dive into Deep Learning

by Aston Zhang, Zachary C. Lipton, Mu Li, and Alex J. Smola

Cambridge University Press, 2023

<http://d2l.ai>

Assumed Knowledge

The course will assume knowledge of the following mathematical topics:

- Linear Algebra
- Probability
- Calculus and Chain Rule

Students should study the relevant sections of the textbook and, if necessary, try to revise these topics on their own during the first few weeks of the course.

Planned Topics

Week 0: Refreshers: Python, Numpy, Matplotlib, Google Colab

Week 1: Neuroanatomy, Perceptrons, Multi-Layer Perceptrons, Backprop

Week 2: Probability, Generalisation & Overfitting, PyTorch

Week 3: Cross Entropy, SoftMax, Weight Decay, Momentum, Hidden Unit Dynamics

Week 4: Convolutional Neural Networks, Image Processing

Week 5: Recurrent Neural Networks (RNN), Long Short-Term Memory Network (LSTM), Gated Recurrent Unit (GRU)

Week 6: Flexibility Week

Week 7: Word Vectors, Language Processing, Large Language Models (LLMs)

Week 8: Reinforcement Learning, TD-learning, Q-learning, Policy Learning, Deep RL

Week 9: Autoencoders, Adversarial Training, Multimodal Learning

Week 10: Generative Artificial Intelligence (GenAI), Review (optional)

Assessments

Assessments will consist of:

- ➔ Assignment (individual): 20%
- ➔ Class Participation (individual): 5%
- ➔ Group Project (group-based): 30%
- ➔ Final Exam (in-person, invigilated): 45%

Due dates:

- ➔ Assignment: Due Week 5
- ➔ Group Project: Due Week 10
- ➔ Final Exam: UNSW Exam Period

Note: In order to pass the course, you must achieve a total mark of at least 50.

Note: Students are expected to form themselves into groups of 5 for the group project in Week 4.

Assignment (individual)

The assignment may involve, for example:

- using code written in PyTorch
- writing your own code
- running experiments and analysing the results

Further details will be provided on the course website.

Participation in Tutorials and Mentoring Sessions

Week 1:	Tutorial Exercises: Tensor Basics	1 mark
Week 2:	Tutorial Exercises: PyTorch, Paper Reading	1 mark
Week 3:	Tutorial Exercises: Paper Reading	1 mark
Week 4:	Tutorial Exercises: Group Formation, Finalizing Project; Finding Dataset(s); Project Planning	1 mark
Week 5:	Tutorial Exercises: Literature Review; Dataset Analysis	1 mark
Week 6:	Flexibility Week	
Week 7:	Implementing baseline method/model; Model training and evaluation; Results analysis; Ideas how existing solution(s) could be improved.	
Week 8:	Implement idea(s) to improve existing work OR comparative analysis OR novel data analysis.	
Week 9:	Discuss main findings of the proposed solution; Discuss strengths and weaknesses of the proposed solution; comparison to other method(s); draft recommendations for future work; draft slides for final presentation.	
Week 10:	Final Project Presentation; Project Submission (Notebook(s), Report, Slides)	

Group Project

The **Group Project** will involve the following:

- ➔ Forming Teams (5 members) in Week 4
- ➔ Try to form a team from within the same Tutorial. However, we do understand that you may want to form a team from other tutorials. In that case, members should be happy to move to different tutorial (change of day/time)
- ➔ Team members can be a mix of undergraduates / postgraduates
- ➔ Choose a Project based on the provided list
- ➔ An assigned Mentor will guide you on the chosen project
- ➔ Discuss project progress with your mentor and seek help throughout the term
- ➔ Each Group need not be present for the entire 2 hour session
- ➔ Project Evaluations in Week 10
- ➔ Deliverables: Source code (Jupyter Notebook), Presentation, Report

Group Project

List of projects will be provided and students have the flexibility to choose one from the list.

Project Title: Insect pest species identification.

Area of Research: Computer Vision

Problem Statement: Insect pest classification plays a crucial role in various domains, including agriculture, pest control, and ecological research. Rapid and accurate identification of insect pests is essential for effective pest management strategies, early detection of invasive species, and preservation of crop yield and quality. However, manual classification of insects based on visual inspection can be time-consuming, error-prone, and challenging, particularly when dealing with conditions in the wild. The goal of this project is to correctly identify the species of insects in an automated manner using advanced artificial intelligence algorithms which has high accuracy, robust to varying environmental conditions, appearance, and deploying these algorithms for real-time monitoring.

Dataset:



Figure 1. Example images of the IP102 dataset. Each image belongs to a different species of insect pests.

The IP102 dataset [1] is a benchmark dataset for insect pests. The details about the dataset can be found in [1] and be downloaded from the URL given below:

Dataset URL: <https://github.com/xpaw95/IP102/tree/master>

Task: To develop an automatic insect recognition system using neural networks and deep learning which provides high accuracy, robust to varying appearance and similarity between different insect species, and faster so that it can be deployed in the real world.

Relevant Papers

- [1]. X. Wu, C. Zhan, Y.-K. Lai, M.-M. Cheng and J. Yang, "IP102: A Large-Scale Benchmark Dataset for Insect Pest Recognition," 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 2019, pp. 8779-8788, doi: 10.1109/CVPR.2019.00899.
<https://www.explore.ieee.org/document/8854351>
https://openaccess.thecvf.com/content_CVPR_2019/papers/Wu_IP102_A_Large-Scale_Benchmark_Dataset_for_Insect_Pest_Recognition_CVPR_2019_paper.pdf
- [2]. A. Selvarani, N. Yuthiraj, and R.C. Whandika, "Large scale pest classification using efficient Convolutional Neural Network with augmentation and regularizers", Computers and Electronics in Agriculture, Vol. 200, Sept 2022.
<https://www.sciencedirect.com/science/article/pii/S0168169922005191>
- [3]. W. Linfeng, L. Yong, L. Jiayao, W. Yunsheng, and X. Shipu, "Based on the multi-scale information sharing network of fine-grained attention for agricultural pest detection", PLOS ONE 18(10):e0286732.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0286732>
- [4]. An J, Du Y, Hong P, Zhang L, Weng X., "Insect recognition based on complementary features from multiple views", Scientific Reports. 2023 Feb;13(1):2966. DOI: 10.1038/s41598-023-29600-1.
<https://www.nature.com/articles/s41598-023-29600-1>
- [5]. S. Kar, J. Nagasubramanian, D. Elango, M. E. Carroll, C. A. Abel, A. Nair, D.S. Mueller, M. E. O'Neill, A. K. Singh, S. Senkar, B. Ganapathy Subramanian, A. Singh, "Self-supervised learning improves classification of agriculturally important insect pests in plants", The Plant Phenome Journal, 6, e20079.
<https://onlinelibrary.wiley.com/doi/full/10.1002/ppj2.20079>

Python, PyTorch and Jupyter Notebooks

We will be using PyTorch for implementing neural networks in this course.

Please install Python and PyTorch on your own laptop. Unless you have a preference for another Python package manager, Anaconda is the recommended one because it installs all dependencies.

```
pytorch.org/get-started/locally/  
www.anaconda.com/
```

Please try to install equal or later versions to these on your own machine:

```
python3 3.11.2      torch 1.13.0  
numpy 1.24.2       sklearn 1.2.1
```

We are going to make heavy use of Jupyter Notebooks for demos and showcasing code examples. Jupyter Notebooks hold text, code, and output, all in the same file, which is very convenient. You can read more about Jupyter Notebooks at: jupyter.org.

Hardware Acceleration for Deep Learning

- ➔ This course will require coding and running tasks that involve heavy numerical computing, such as multiplying large numbers of matrices. Although you can follow along with much of the content using a standard personal computer or laptop with PyTorch installed on it. However, we anticipate that training deep learning algorithms from scratch or running a pretrained network, and doing assessments will require a CUDA-enabled GPU machine.
- ➔ Note that currently only NVIDIA's graphics cards support CUDA and cuDNN libraries, and hence can be used for accelerated training of deep learning algorithms.
- ➔ Cloud based options: Google Colab

Week 1 (Tutorial)

- Tutorials available on WebCMS3 → Tutorials
- Tutorials will be available on Friday (for the following week's tutorial)
- **Checkpoint #1**: Python Exercises and Tensor Basics

In the last 40 minutes of the Week 1 Tutorial, you will work on refreshing Python and learning basics of Tensors. Follow the links (in the Tutorial Questions) to the **Python Refresher**, **Python Exercises**, **Numpy Refresher**, **Matplotlib**, **PyTorch Tensors** and **Google Colab**.

Copy-paste and run the provided code fragments, and complete the Exercises.

Show your work to your tutor as a checkpoint for this tutorial.

Week 2 (Tutorial)

- In the **Week 2 Tutorial**, as well as the tutorial questions, **we will be reading and having a discussion about a research paper.**
- The goal is to analyse the paper from a variety of different vantage points. During these tutorial sessions, students will be forming groups of 4 to 5, **each student with assigned role will provide one particular perspective.**
- Everyone in the class should participate.
- Paper discussions will be moderated by tutors.

The Roles

- Archaeologist
- Social Impact Assessor
- Industry Practitioner
- Researcher
- Scientific Peer Reviewer

Inspired by: Colin Raffel and Alec Jacobsen's
Role-playing Paper Reading Seminars

Plagiarism

- Plagiarism is taken seriously by UNSW/CSE and treated as Academic Misconduct. ALL work submitted for assessment must be your own work.
- For an individual assignment, collaborative work in the form of “think tanking” is encouraged, but students are not allowed to derive code together as a group during such discussions. In the case of a group assignment, code must not be obtained from outside the group.
- Plagiarism detection software may be used on submitted work.
- Academic Integrity and Plagiarism:
<https://student.unsw.edu.au/plagiarism>

Related Courses

- COMP3411/9414 Artificial Intelligence
- COMP9417 Machine Learning and Data Mining
- COMP9418 Advanced Topics in Statistical Machine Learning
- COMP4418 Knowledge Representation and Reasoning
- COMP9491 Applied Artificial Intelligence
- COMP9517 Machine Vision
- COMP3431 Robotic Software Architecture
- COMP9727 Recommender Systems
- COMP6713 Natural Language Processing
- 4th Year Thesis topics

Deep Learning: Advances, Threats and Challenges

→ Recent Advances:

- Game Playing
- Medical Research and Development
- Text Generation, ChatBots, Code Generation
- Image, Voice, Song, Video Generation
- Object Detection, Image Segmentation and Captioning, Multimodal AI
- Agents, Decision-Making AI

→ Threats and Challenges:

- Technological Unemployment
- Loss of Human Meaning / Purpose / Agency
- Energy Consumption
- Existential Threats

Will Technology Create or Destroy More Jobs?

